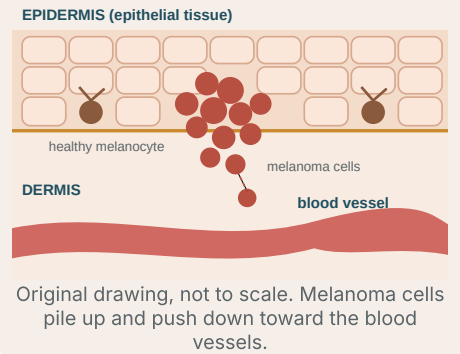


1 MELANOMA

When the skin's pigment cells stop taking orders

Melanoma is a skin cancer that starts in **melanocytes**, the cells that make the brown pigment melanin. It is less common than other skin cancers, but it is the one most likely to spread to other organs. Caught early, it is one of the most curable cancers there is.



2 Signs, and how serious it is

DESCRIPTION

Most melanomas show up as a new mole or a mole that changes. Doctors teach the ABCDEs:

A

Asymmetry:
halves
don't match

B

Border:
ragged or
blurred

C

Color:
uneven,
several
shades

D

Diameter:
wider than
6 mm

E

Evolving:
changing
over time

Itching, oozing or bleeding are warning signs too. How serious it is depends on how far it has spread.

3 What causes it

CAUSES

Ultraviolet (UV) light from the sun and tanning beds damages the instructions in the nucleus of skin cells. Blistering sunburns in childhood raise the risk most.

Risk is also higher with fair skin that freckles, many moles, a weakened immune system, or a family history of melanoma.

Still being studied: some melanomas grow where the sun rarely reaches, like the soles of the feet. Scientists hypothesize that (1) inherited changes in a family's instructions and (2) chemicals such as some solvents can start melanoma without much sun.

4 5 6 7 Where it fits in the body's levels of organization

CELLS · TISSUE · ORGAN · SYSTEM

CELL

Melanocyte

makes melanin, the pigment that shades skin from UV

TISSUE

Epithelial tissue

melanocytes sit in the bottom layer of the epidermis

ORGAN

Skin

if it spreads: lymph nodes, lungs, liver, brain

ORGAN SYSTEM

Integumentary system

if it spreads: lymphatic, respiratory, nervous systems

8 How it upsets homeostasis

HOMEOSTASIS

HEALTHY SKIN KEEPS...
germs and water on the right side of a barrier, and helps hold body temperature steady.

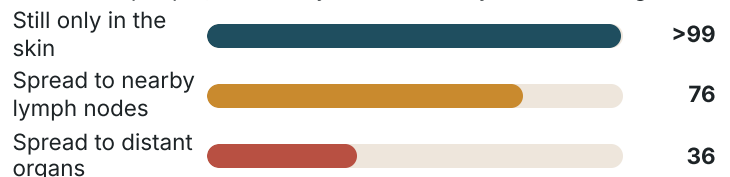
MELANOMA...
breaks the barrier with sores that bleed, and crowds out healthy cells that need the same blood supply.

After it spreads, tumors in the lungs make it hard to keep blood oxygen steady, and tumors in the liver or brain stop those organs from doing their jobs.

Why catching it early matters

SEVERITY

Out of 100 people, how many are alive five years after diagnosis:



People diagnosed 2015–2021 (American Cancer Society, 2026).

9 How it is treated so the body can return to balance

TREATMENT

Surgery

Cut out the melanoma with a border of healthy skin. Early melanoma is often cured this way, and the skin heals and closes the barrier again.

Immunotherapy

Medicines that help the body's own defense cells recognize melanoma cells and destroy them.

Targeted therapy

Drugs aimed at a changed protein that keeps certain melanoma cells dividing.

Radiation & chemotherapy

Used when melanoma has spread, to shrink tumors so organs like the lungs or brain can work again.

Prevention: shade, sunscreen of SPF 30 or higher, no tanning beds, and checking your skin for ABCDE changes.

References

RESOURCES · APA

American Academy of Dermatology Association. (n.d.). *Melanoma: Signs and symptoms*. <https://www.aad.org/public/diseases/skin-cancer/types/common/melanoma/symptoms>

American Cancer Society. (2026, August 12). *Survival rates for melanoma skin cancer*. <https://www.cancer.org/cancer/types/melanoma-skin-cancer/detection-diagnosis-staging/survival-rates-for-melanoma-skin-cancer-by-stage.html>

National Cancer Institute. (2025, May 16). *Melanoma treatment (PDQ®)—patient version*. <https://www.cancer.gov/types/skin/patient/melanoma-treatment-pdq>